# An Open Software Development-based Ecosystem of R Packages for Proteomics Data Analysis

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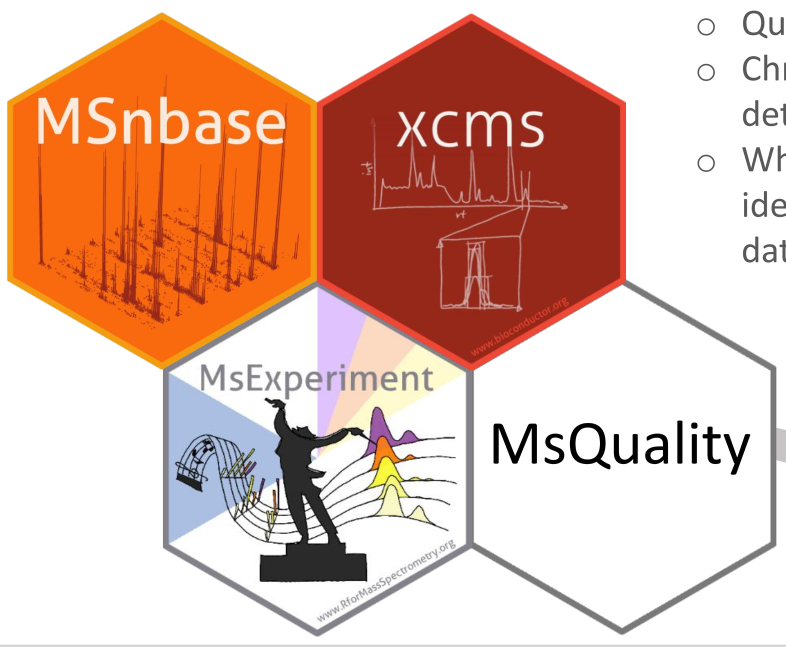
## In a nutshell

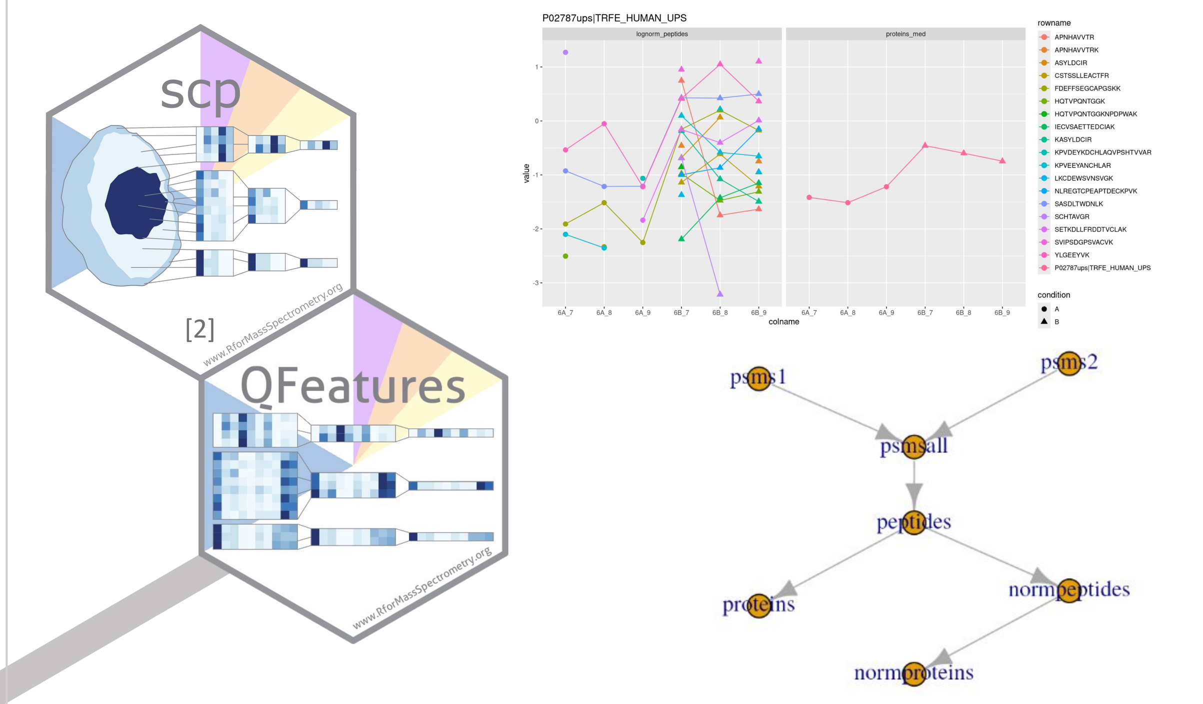
- Powerful and flexible infrastructure for efficient analysis of large-scale MS data and quantitative proteomics.
- Open collaborative development of software for MS data analysis:
  - Well documented and thoroughly tested
  - Long-term support and maintenance
- Modular package ecosystem to allow creation of custom analysis workflows
- Tightly integrated into Bioconductor

## Interoperability



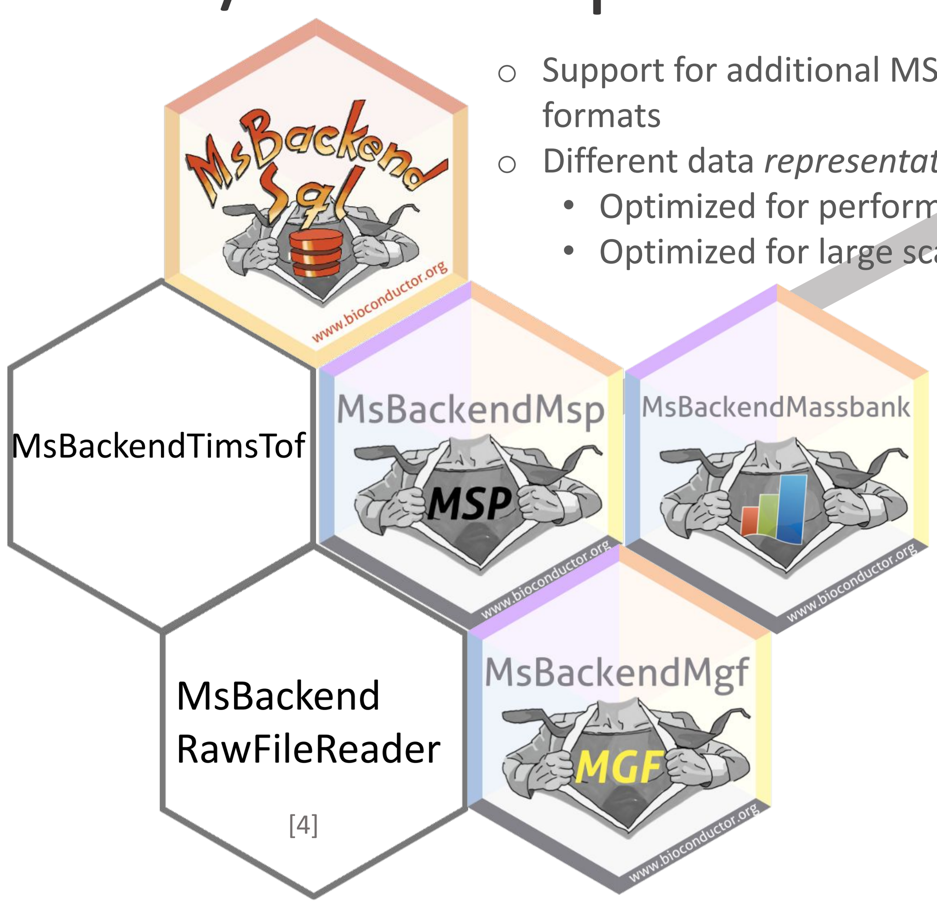

- MSnbase package [0]
- Quality assessment [1]
- Chromatographic peak detection and alignment
- Whole experiment (raw + identification + quantitative data) integration


## Quantitative proteomics



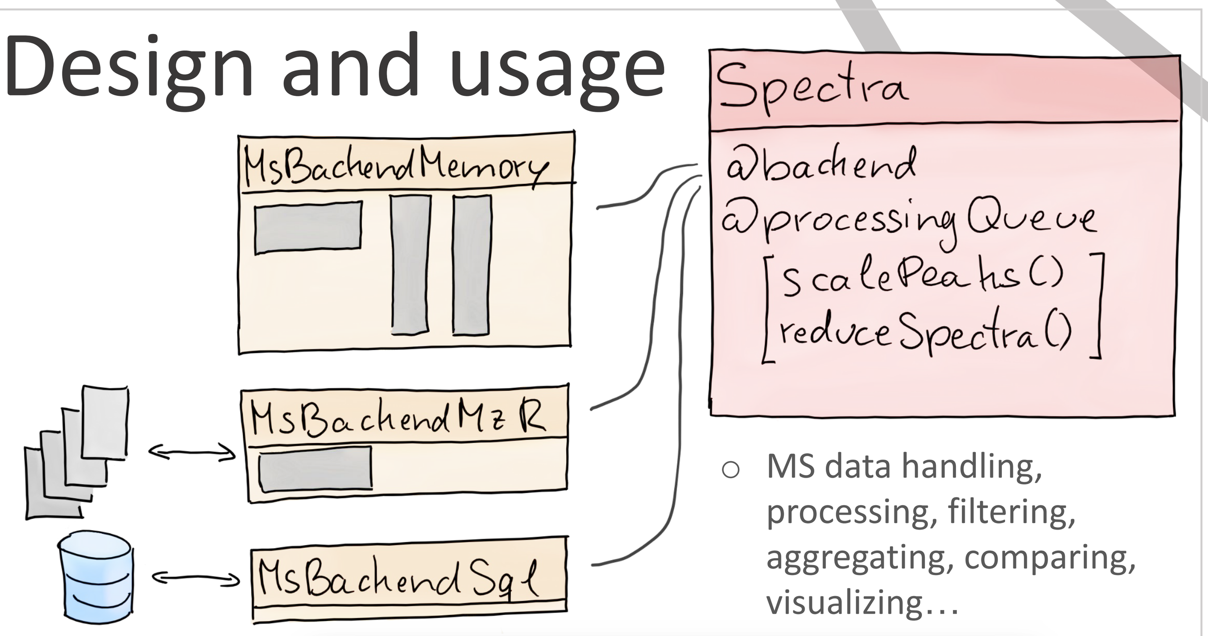

- Import quantitative data (see also the **QFeaturesGUI** poster)
- Quantitative data handling and processing: feature aggregation, missing data filtering and imputation, normalisation.
- Exploratory data and statistical analyses [3]
- Optimised for bulk and single-cell proteomics (see **scp** and **scpGUI** posters)
- Build upon Bioconductor's `SummarizedExperiment` and `MultiAssayExperiment` infrastructure

## Data I/O and representation




- Support for additional MS data file formats
- Different data *representations* :
  - Optimized for performance
  - Optimized for large scale data

## Design and usage

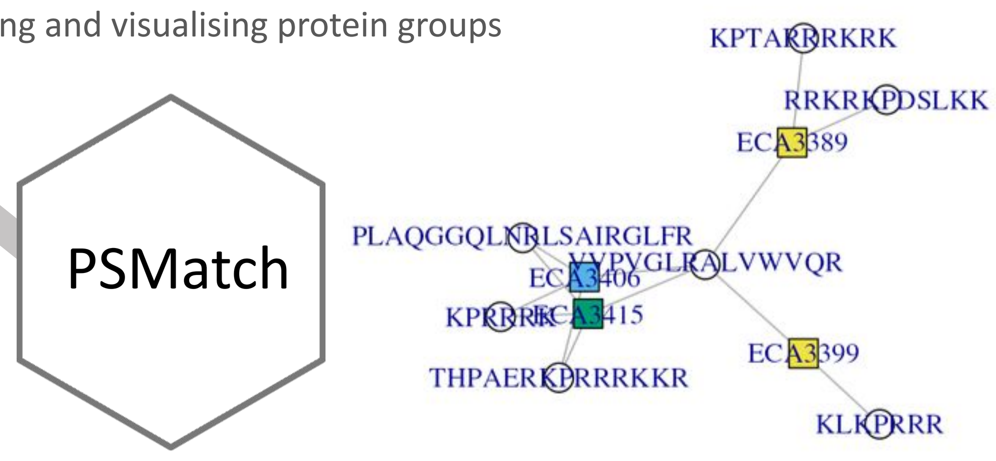



- MS data handling, processing, filtering, aggregating, comparing, visualizing…

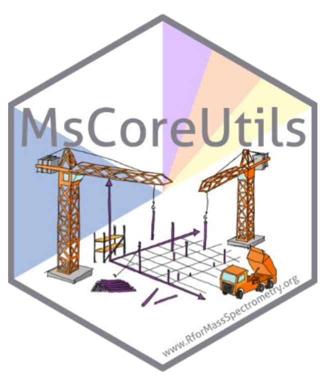
```
> filterMsLevel(sps, msLevel = 1L) |>
+   scalePeaks() |>
+   reduceSpectra()
```

## Peptide identification

- Peptide-Spetrum Matches data processing
- MS2 Fragment ions and spectra annotations
- Exploring and visualising protein groups




## Core *low-level* functions




- For raw MS and quantitative data processing
- Spectra similarity scores
- Peak alignment
- Normalisation
- Imputation
- Aggregation
- …

## Outlook

- Continue integration of R/Bioconductor packages and external software
- Emphasis on seamless integration to improve user experience
- Open for contributions from the community -> reach out via github issue or email
- Shared infrastructure and development with the metabolomics community within *R for Mass Spectrometry* initiative

## References

[0] Gatto L, Gibb S, Rainer J *MSnbase, efficient and elegant R-based processing and visualisation of raw mass spectrometry data* J. Proteome Res. 2021, 20, 1, 1063–1069, doi:10.1021/acs.jproteome.0c00313 (preprint bioRxiv:10.1101/2020.04.29.067868).
[1] Naake T et al. *MsQuality – an interoperable open-source package for the calculation of standardized quality metrics of mass spectrometry data*. Bioinformatics 2023. doi.org:10.1093/bioinformatics/btad618
[2] Vanderaa C and Gatto L *The current state of single-cell proteomics data analysis* (2023), Current Protocols 3 (1):e658, doi:10.1002/cpz1.658 (preprint: arXiv:2210.01020).
[3] Vanderaa C and Gatto L *scplainer: using linear models to understand mass spectrometry-based single-cell proteomics data* (2024) bioRxiv 2023.12.14.571792; doi:10.1101/2023.12.14.571792.
[4] Kockmann T et al. *The rawrr R Package: Direct Access to Orbitrap Data and Beyond*. Journal of Proteome Research 2021. doi:10.1021/acs.jproteome.0c00866